

Fig. 2.5 The error in energy in a numerically computed trajectory (Verlet method) with stepsize $h = 0.00001$, for the anisotropic oscillator, showing that it remains bounded by 1.5×10^{-11} in very long runs

trajectories for the anisotropic oscillator shown in Fig. 1.25, the energy errors (see Fig. 2.5) remain bounded and small, suggesting that we may have some confidence in the numerical results presented. This example also provides additional evidence for the lack of "secular growth"[6] of the energy error in Verlet simulations even in a relatively large number (10^8) of timesteps.

2.2.4 *First Integrals and Their Preservation Under Discretization*

Recall that the condition for a given function $I : \mathbb{R}^m \to \mathbb{R}$ to be a first integral is that

$$\nabla I(z) \cdot f(z) = \sum_{j=1}^{m} \frac{\partial I}{\partial z_j}(z) f_j(z) \equiv 0.$$

It is important in this definition that this is an equivalence that holds *everywhere* (or at least in some open set in $\mathbb{R}^m$), so the statement is not just that $I(z(t)) = I(z(0))$ for some particular trajectory, but, moreover, I is conserved for all nearby initial conditions. The flow map preserves the first integral, thus $I(\mathscr{F}_t(z)) \equiv I(z)$. For example, in a Hamiltonian system, the flow map conserves the energy:

$$H \circ \mathscr{F}_t = H.$$

[6]The term "secular growth" in this context is a reference to the long-term growth of perturbations in celestial mechanics. For example, the precession of the Earth's polar axis occurs on a long period relative to its orbital motion and much longer period than its rotation, and so may be classed as a secular motion. In the context of molecular simulations, we use this to refer to accumulation of drift that takes the system steadily away from the energy surface.